

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

PRACTICAL IV(Task)

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Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:-

```
P_Python = 0.20
P_Java = 0.80
P_Placed_Python = 0.90
P_Placed_Java = 0.40
P_Placed = (P_Placed_Python * P_Python) + (P_Placed_Java * P_Java)
P_Python_Placed = (P_Placed_Python * P_Python) / P_Placed
print("Probability student knows Python:", P_Python)
print("Probability student knows Java:", P_Java)
print("Probability placed student knows Python:",
      round(P_Python_Placed * 100, 2), "%")
```

Output:-

```
Probability student knows Python: 0.2
Probability student knows Java: 0.8
Probability placed student knows Python: 36.0 %
```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

Code:-

```
P_Blue = 0.15
P_Green = 0.85
P_SaysBlue_Blue = 0.80
P_SaysBlue_Green = 0.20
P_SaysBlue = (P_SaysBlue_Blue * P_Blue) + (P_SaysBlue_Green * P_Green)
P_Blue_SaysBlue = (P_SaysBlue_Blue * P_Blue) / P_SaysBlue
print("Probability taxi was actually Blue:",
      round(P_Blue_SaysBlue * 100, 2), "%")
```

Output:-

```
Probability taxi was actually Blue: 41.38 %
```